

Materials Providing INVISIBILITY TO ELECTRONICS



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Though development and use of invisibility cloaks may not have been possible till now for large-scale objects, the technology towards invisibility based on the exotic properties exhibited by meta-materials is now well understood by scientists, engineers and technologists. So we are no longer unaware of the use of invisibility in electronics.

As components on computer chips get smaller, we have to come up with strategies to control electron transport, and use of invisibility characteristics of materials in electronics might be one useful approach. The concept could also lead to a new kind of switches for electronic devices. The switch could operate by toggling between transparent and opaque states of electrons, thus turning a flow of these on and off. This article briefly summarises the developments towards the applications of the concept of cloaking to the domain of electrons that is giving rise to more useful, efficient,

smart and lightweight electronic devices. The new invisibility cloak combines meta-materials and leads fancy electronics to be thinner, lighter and invisible.

Invisibility fundamentals

The last few years have seen a lot of research in invisibility cloaks. These cloaks are mostly based on meta-materials—special, man-made materials that bend radiation in ways that should not technically be possible—allowing for cloaking devices that bend radiation around an object, hiding it from view.

The problem with these cloaks is that meta-materials are tuned to a very specific frequency. So while that specific frequency passes around the object, every other frequency scatters off the cloaks.

In a beautiful twist of irony, most invisibility cloaks actually create more scattered light, making the cloaked object stand out more than if it was just standing there uncloaked. This is a fundamental issue of passive invisibility cloaks, and

*Fig. 1: What
meta-material
looks like*

